

Geometrical Problems

Pythagoras' Theorem. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner applies Pythagoras' Theorem in richer settings — isosceles triangles, distances between points, and multi-step figures — where the right triangle must first be **found or constructed**. The lesson joins earlier skills: spotting the right angle, deciding add-versus-subtract, computing carefully, then interpreting the result in context.

Driving Question: *How can a hidden right triangle unlock a geometry problem?*

The mathematics, in plain terms

Many geometry problems contain a right triangle that is not drawn for you. A common example: dropping a perpendicular from the apex of an isosceles triangle to the midpoint of the base splits it into two identical right triangles, and that perpendicular meets the base at a right angle (it is the perpendicular bisector of the base). Once a right triangle is visible, the learner decides whether the unknown is the hypotenuse (add the squares) or a shorter side (subtract), computes, and interprets the answer back in the original figure.

Find the triangle first, then decide, then interpret

The habit worth building is a fixed order: **find** a usable right triangle (drawing one line if needed), **decide** whether the unknown is the hypotenuse or a shorter side, **calculate**, then **interpret** the number in the original problem. Learners most often stall at the first and last steps — spotting the triangle, and remembering that the number they found answers a real question about the figure. A quick sketch, and the question “what does this length mean here?”, address both.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	A hidden right triangle connects an everyday shape to Pythagoras' Theorem; the answer connects back to the original problem.	~8 min
Movement	Constructing a helpful line to reveal the right triangle, then working through it.	~22 min
Reflection	Checking: is this really a right angle? Does my answer make sense in the original figure?	~4 min

Creativity	Building a worked card for a problem where the triangle is hidden.	~3 min
Rest	The pause to sketch and see the structure before calculating.	~3 min
Nutrition	Intellectual (Mode B): feeds the habit of finding structure in unfamiliar situations.	~5 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Not spotting the right triangle

The learner tries to use side lengths directly. Repair: “Where is there a right angle in this figure — or where could you draw one line to make one?”

Choosing the wrong operation

Repair: “Is the length you want the longest side of your right triangle, or a shorter one? Longest means add; shorter means subtract.”

In an isosceles triangle, not using the perpendicular from the apex

Repair: “What happens if you drop a straight line from the top to the middle of the base? What angle does it make, and what two triangles appear?”

Finding a length but not answering the question

Repair: “You have a number — now, what does it measure in the original problem? Does it answer what was asked?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: choosing whether to add or subtract, square roots, and spotting the hidden right angle. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Guide the finding and the interpreting, not the arithmetic:

- “Can you see a right angle anywhere? If not, is there one line you could draw to make one?”

- “In your right triangle, which side is the longest? Is that the one you want?”
- “Add or subtract the squares? Why?”
- “What does your answer mean back in the original shape?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“This looks harder than the earlier lessons.”

It uses the same theorem — the new skill is *finding* the right triangle. Once it is visible, the method is exactly the same as before.

“Do they need to remember the isosceles trick?”

The useful idea is general: a line from the apex to the midpoint of the base is perpendicular. Prompting them to try drawing it is better than naming a trick to memorise.

“Their answer is a decimal — is that fine?”

Yes. Real measurements rarely come out whole; a sensible rounded answer with working shown is strong.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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